GCSE CHEMISTRY

Higher Tier – Paper 1 style

Topic 4.4: Chemical changes

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 140 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **134**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 134

0 1 A student adds four metals to dilute hydrochloric acid and records what she sees. The results are shown.

0 1 . 1 Put the four metals in order of reactivity. Start with the most reactive. **[2 marks]**

Metal	Observation with dilute hydrochloric acid
magnesium	very fast fizzing, test tube gets warm
zinc	steady stream of bubbles
iron	a few bubbles, very slow
copper	no bubbles

0 1 . 2 Name the gas produced when a metal reacts with an acid and describe the test for the gas. **[2 marks]**

0 1 . 3 Write a word equation and a balanced symbol equation for the reaction of magnesium with hydrochloric acid. **[3 marks]**

0 1 . 4 Calcium reacts with cold water. Write a balanced symbol equation for this reaction. **[2 marks]**

0 1 . 5 Magnesium burns in oxygen. Write a balanced symbol equation for the reaction. **[2 marks]**

0 1 . 6 The more reactive a metal is, the more easily it forms positive ions. Explain what this means for the atoms of a metal. **[2 marks]**

0 1 . 7 Gold is found in the Earth as the uncombined metal. Explain why. **[1 mark]**

0 2 A student adds pieces of four metals to solutions of metal sulfates. The table shows whether a reaction happened (✓) or not (✗). A dash (–) means no experiment.

0 2 . 1 Put the four metals in order of reactivity. Start with the most reactive. **[2 marks]**

Metal added	magnesium sulfate	zinc sulfate	iron(II) sulfate	copper(II) sulfate
magnesium	–	✓	✓	✓
zinc	✗	–	✓	✓
iron	✗	✗	–	✓
copper	✗	✗	✗	–

0 2 . 2 Zinc is added to copper(II) sulfate solution. Describe **two** things the student sees. **[2 marks]**

0 2 . 3 Write a word equation and a symbol equation for the reaction of zinc with copper(II) sulfate. **[2 marks]**

0 2 . 4 Explain why zinc displaces copper from copper(II) sulfate solution. **[1 mark]**

0 2 . 5 The ionic equation for the reaction is $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ **[3 marks]**
Which species is oxidised and which is reduced? Give a reason for each.

0 2 . 6 Name the spectator ions in the reaction between zinc and copper(II) sulfate solution. **[1 mark]**

0 3 This question is about the extraction of metals.

0 3 . 1 Iron is extracted from iron oxide by heating with carbon. Give the **name** of the type of reaction that happens to the iron oxide, and explain your answer. **[2 marks]**

0 3 . 2 Balance the equation for the extraction of iron. **[2 marks]**
___ Fe_2O_3 + ___ C → ___ Fe + ___ CO_2

0 3 . 3 Which substance is oxidised in this reaction? Give a reason. **[2 marks]**

0 3 . 4 Aluminium cannot be extracted from aluminium oxide by heating with carbon. Explain why. **[2 marks]**

0 3 . 5 Metals more reactive than carbon are extracted by electrolysis. Give **one** reason why electrolysis makes the metal expensive. **[1 mark]**

0 4 This question is about oxidation and reduction.

0 4 . 1 Describe oxidation and reduction in terms of electrons. **[2 marks]**

0 4 . 2 Magnesium reacts with copper oxide: $Mg + CuO \rightarrow MgO + Cu$ **[3 marks]**
Which substance is oxidised and which is reduced? Explain your answers in terms of oxygen.

0 4 . 3 Complete the half equations. State whether each is oxidation or reduction. **[2 marks]**

$Mg \rightarrow Mg^{2+} +$ ____
$Cu^{2+} + 2e^{-} \rightarrow$ ____

0 4 . 4 Iron reacts with copper(II) sulfate solution. Explain, in terms of electrons, why this is a redox reaction. **[3 marks]**

0 5 This question is about the reactions of acids.

0 5 . 1 Complete the word equations. **[4 marks]**

Reaction
sulfuric acid + copper oxide → _____ + _____
hydrochloric acid + sodium hydroxide → _____ + _____
nitric acid + calcium carbonate → _____ + _____ + _____
sulfuric acid + zinc → _____ + _____

0 5 . 2 Write a balanced symbol equation for the reaction of magnesium oxide with sulfuric acid. **[2 marks]**

0 5 . 3 Calcium carbonate reacts with hydrochloric acid and carbon dioxide is made. **[2 marks]**
Describe what you would see and how you would test for carbon dioxide.

0 5 . 4 Write the ionic equation for neutralisation. **[1 mark]**

0 5 . 5 What is the difference between a base and an alkali? **[2 marks]**

0 6 A student prepares crystals of copper(II) sulfate from copper(II) oxide (an insoluble base) and dilute sulfuric acid.

0 6 . 1 *Describe how the student could make pure, dry crystals of copper(II) sulfate. **[6 marks]**

0 6 . 2 The student adds the copper oxide in excess. Why? **[1 mark]**

0 6 . 3 How can the student tell that the copper oxide is in excess? **[1 mark]**

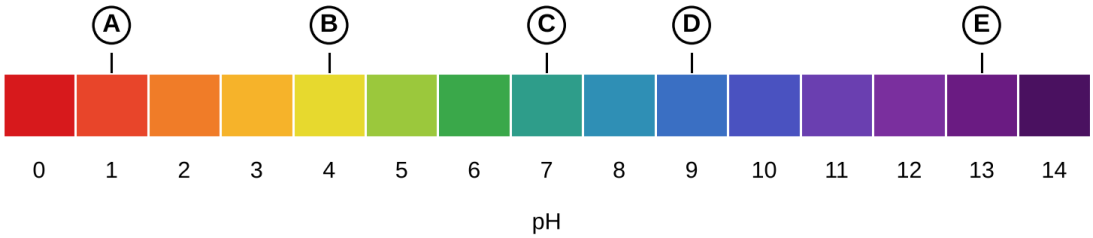
0 6 . 4 The acid is warmed before the copper oxide is added. Suggest why. **[1 mark]**

0 6 . 5 The student does not evaporate all of the water from the filtrate. Suggest why. **[1 mark]**

0 6 . 6 Name the salt made when zinc carbonate reacts with hydrochloric acid. **[1 mark]**

0 6 . 7 This method cannot be used to make sodium chloride from sodium hydroxide and hydrochloric acid. **[2 marks]**
Explain why, and suggest how sodium chloride can be made.

0 7 The pH scale is used to show how acidic or alkaline a solution is. The diagram shows the pH of five solutions, **A** to **E**.



0 7 . 1 Which solution is: (i) a strong acid, (ii) a weak acid, (iii) neutral, (iv) strongly alkaline? Write letters. **[4 marks]**

0 7 . 2 Describe the colour of universal indicator in solution **A** and in solution **E**. **[2 marks]**

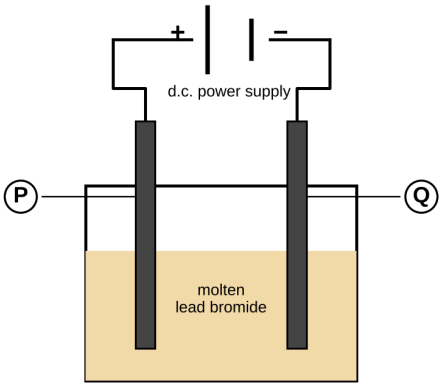
0 7 . 3 Hydrochloric acid is a strong acid. Ethanoic acid is a weak acid. **[2 marks]**
Explain the difference between a strong acid and a weak acid.

0 7 . 4 Solution X has a pH of 3. Solution Y has a pH of 5. **[2 marks]**
How many times greater is the concentration of hydrogen ions in X than in Y?

0 7 . 5 Explain the difference between a **concentrated** acid and a **strong** acid. **[2 marks]**

0 7 . 6 Magnesium reacts more slowly with 1 mol/dm^3 ethanoic acid than with 1 mol/dm^3 hydrochloric acid. **[2 marks]**
Explain why.

0 8 The diagram shows apparatus used to electrolyse molten lead bromide using graphite electrodes.



0 8 . 1 Why must the lead bromide be molten for electrolysis to happen? **[2 marks]**

0 8 . 2 Which electrode is the anode? Write **P** or **Q**. **[1 mark]**

0 8 . 3 Name the product formed at each electrode. **[2 marks]**

0 8 . 4 Explain why lead is produced at the negative electrode. **[3 marks]**

0 8 . 5 Write the half equation for the reaction at the anode. **[2 marks]**

0 8 . 6 Suggest why graphite is used for the electrodes. **[1 mark]**

0 8 . 7 Predict the products when molten sodium chloride is electrolysed. **[2 marks]**

0 9 A student electrolyses four different aqueous solutions using inert (graphite) electrodes.

0 9 . 1 Complete the table to show the product at each electrode. **[4 marks]**

Solution	Product at the cathode (−)	Product at the anode (+)
copper(II) chloride		
sodium chloride		
sodium sulfate		
copper(II) sulfate		

0 9 . 2 In the electrolysis of sodium chloride solution, hydrogen forms at the cathode and not sodium. **[2 marks]**
Explain why.

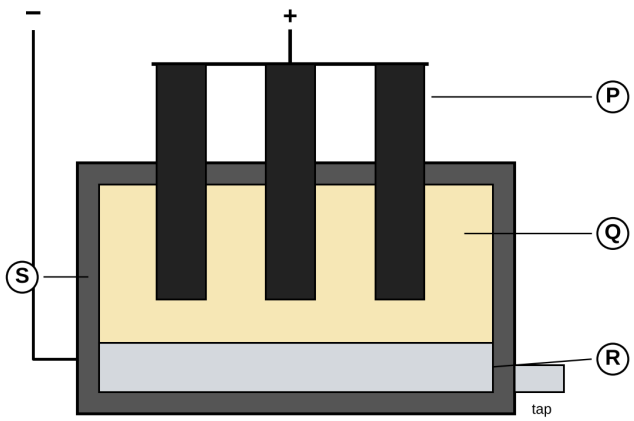
0 9 . 3 Describe a test for each gas: chlorine, hydrogen and oxygen. **[3 marks]**

0 9 . 4 In the electrolysis of copper(II) sulfate solution, oxygen forms at the anode. **[2 marks]**
Complete the half equation.
$4OH^- \rightarrow O_2 + 2H_2O +$ ____

0 9 . 5 Describe how the student could collect and identify the gases made at each electrode. **[3 marks]**

0 9 . 6 During the electrolysis of copper(II) chloride solution, the blue-green colour fades. **[2 marks]**
Explain why.

1 0 Aluminium is extracted from aluminium oxide by electrolysis. The diagram shows the electrolysis cell.



1 0 . 1 Name the parts labelled **P** and **S**. **[2 marks]**

1 0 . 2 The aluminium oxide is dissolved in molten cryolite. Give **one** reason why. **[2 marks]**

1 0 . 3 Name the product formed at the cathode and the product formed at the anode. **[2 marks]**

1 0 . 4 The anodes have to be replaced regularly. Explain why. **[2 marks]**

10.5 Write the half equations for the reactions at the cathode and the anode. **[2 marks]**

10.6 The extraction of aluminium is expensive. Give **one** reason why. **[1 mark]**

11 Stretch questions (grade 7–9).

11.1 *Describe and explain what happens when molten lead bromide is electrolysed using graphite electrodes. **[6 marks]**
Include the products, the half equations and the type of reaction at each electrode.

1 1 . 2 *Explain how the reactivity series can be used to predict (i) which metals will displace others from solution and (ii) how a metal is extracted from its ore. Use examples in your answer. **[6 marks]**

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.4: Chemical changes

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	magnesium, zinc, iron, copper ; (all four in the correct order for 2 marks; 3 in the correct relative order for 1 mark) ;	More bubbles / faster fizzing = more reactive. Copper does not react.	2	AO3 4.4.1.1
01.2	hydrogen ; lit splint makes a squeaky pop ;		2	AO1 4.4.1.1
01.3	magnesium + hydrochloric acid → magnesium chloride + hydrogen ; $Mg + 2HCl \rightarrow MgCl_2 + H_2$; balanced ;	Mark 2 is for correct formulae and mark 3 for the equation balanced.	3	AO1 4.4.1.1
01.4	$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$; correct formulae and balanced ;	One mark for correct formulae, one for balancing.	2	AO2 4.4.1.1
01.5	$2Mg + O_2 \rightarrow 2MgO$; correct formulae and balanced ;		2	AO1 4.4.1.1
01.6	more reactive metals lose (outer) electrons more easily ; to form positive ions ;		2	AO2 4.4.1.1
01.7	gold is very unreactive / does not react with oxygen or water		1	AO1 4.4.1.2
02.1	magnesium, zinc, iron, copper ; (all four in correct order for 2 marks; three correct for 1) ;	A metal displaces any metal less reactive than itself from solution.	2	AO3 4.4.1.2
02.2	a reddish-brown / pink solid forms (copper) ; the blue solution fades / becomes paler (colourless) ;	Do not accept 'copper turns red' or 'the zinc turns brown' unless it is a coating on the zinc.	2	AO2 4.4.1.2
02.3	zinc + copper(II) sulfate → zinc sulfate + copper ; $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$;		2	AO1 4.4.1.2
02.4	zinc is more reactive than copper		1	AO1 4.4.1.2
02.5	zinc (atoms) oxidised ; because they lose electrons ; Cu^{2+} ions are reduced because they gain electrons ;	OIL RIG: Oxidation Is Loss (of electrons), Reduction Is Gain.	3	AO2 4.4.2.1
02.6	sulfate ions / SO_4^{2-}		1	AO2 4.4.2.1
03.1	reduction ; the iron oxide loses oxygen ;		2	AO1 4.4.2.2
03.2	$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$; (1 mark for iron balanced, 1 mark for the complete balanced equation) ;		2	AO2 4.4.1.3
03.3	carbon ; it gains oxygen (to form carbon dioxide) ;		2	AO2 4.4.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	aluminium is more reactive than carbon ; so carbon cannot displace / remove oxygen from aluminium oxide ;	Carbon only displaces metals less reactive than carbon.	2	AO2 4.4.1.3
03.5	uses a lot of electricity / energy ; the ore has to be melted (high temperature) ;	Any one.	1	AO1 4.4.1.3
04.1	oxidation is loss of electrons ; reduction is gain of electrons ;	OIL RIG.	2	AO1 4.4.2.1
04.2	magnesium is oxidised ; it gains oxygen ; copper oxide is reduced because it loses oxygen ;		3	AO2 4.4.2.1
04.3	$2e^-$: oxidation ; Cu: reduction ;	One mark for each half equation completed and labelled correctly.	2	AO2 4.4.3.2
04.4	iron atoms lose electrons to form Fe^{2+} ions (oxidation) ; copper ions gain electrons to form copper atoms (reduction) ; both oxidation and reduction happen in the same reaction ;		3	AO3 4.4.2.1
05.1	copper sulfate + water ; sodium chloride + water ; calcium nitrate + water + carbon dioxide ; zinc sulfate + hydrogen ;	Hydrochloric acid → chlorides; sulfuric acid → sulfates; nitric acid → nitrates. One mark for each correct equation.	4	AO1 4.4.3.1
05.2	$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$; correct formulae and balanced ;		2	AO2 4.4.3.1
05.3	fizzing / bubbles / solid disappears ; bubble the gas through limewater, which turns cloudy / milky ;		2	AO1 4.4.3.1
05.4	$H^+ + OH^- \rightarrow H_2O$		1	AO1 4.4.3.1
05.5	a base is a substance that neutralises an acid (metal oxide / hydroxide) ; an alkali is a base that dissolves in water / forms hydroxide ions, $OH^-(aq)$;		2	AO1 4.4.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • warm the dilute sulfuric acid (in a beaker) • add copper(II) oxide in small amounts, stirring, until no more dissolves (excess) • filter to remove the excess copper(II) oxide • heat the filtrate in an evaporating basin to evaporate some of the water (crystallisation point) • leave the solution to cool and crystallise (slowly) • filter off the crystals and dry them between filter paper / in a warm oven	Level of response question.	6	AO1 4.4.3.1
06.2	to make sure all the acid reacts / so no acid is left in the solution		1	AO1 4.4.3.1
06.3	some solid copper oxide is left over / no more dissolves		1	AO1 4.4.3.1
06.4	to speed up the reaction		1	AO3 4.4.3.1
06.5	to allow large crystals to form on cooling / evaporating to dryness would leave a powder / crystals may be damaged by overheating		1	AO3 4.4.3.1
06.6	zinc chloride		1	AO1 4.4.3.1
06.7	sodium hydroxide is soluble so the excess cannot be filtered off ; use a titration to find the exact volumes of acid and alkali (then evaporate) ;		2	AO3 4.4.3.1
07.1	strong acid: A ; weak acid: B ; neutral: C ; strongly alkaline: E ;		4	AO2 4.4.3.2
07.2	A: red ; E: purple / dark blue ;	Allow orange-red for A and blue-purple for E.	2	AO1 4.4.3.2
07.3	a strong acid is completely ionised in water ; a weak acid is only partly ionised ;	Allow 'completely dissociated'.	2	AO1 4.4.3.3
07.4	pH differs by 2 ; each unit is a factor of 10 so 10 × 10 = 100 times ;		2	AO2 4.4.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	concentrated = a large amount of acid in a given volume of solution ; strong = fully ionised, whatever the concentration ;	Common error: using 'concentrated' and 'strong' as if they meant the same.	2	AO3 4.4.3.3
07.6	ethanoic acid is a weak acid (only partly ionised) ; lower concentration of H^+ ions so fewer collisions / slower reaction ;		2	AO3 4.4.3.3
08.1	the ions are free to move (in a liquid) ; so they can carry the charge / move to the electrodes ;	In the solid the ions are held in fixed positions.	2	AO1 4.4.4.1
08.2	P	The anode is connected to the positive terminal.	1	AO2 4.4.4.1
08.3	P: bromine ; Q: lead ;	Allow Br_2 and Pb.	2	AO1 4.4.4.1
08.4	lead ions (Pb^{2+}) are positive so are attracted to the negative electrode ; they gain electrons ; this is reduction (to form lead atoms) ;	Positive ions move to the negative electrode (cathode).	3	AO2 4.4.4.1
08.5	$2Br^- \rightarrow Br_2 + 2e^-$; correct formulae and balanced / electrons on the right ;	Allow $2Br^- - 2e^- \rightarrow Br_2$.	2	AO2 4.4.4.2
08.6	it conducts electricity / is inert (unreactive) / has a high melting point		1	AO3 4.4.4.1
08.7	sodium at the cathode ; chlorine at the anode ;		2	AO2 4.4.4.1
09.1	copper(II) chloride: copper, chlorine ; sodium chloride: hydrogen, chlorine ; sodium sulfate: hydrogen, oxygen ; copper(II) sulfate: copper, oxygen ;	Cathode: the metal if it is less reactive than hydrogen, otherwise hydrogen. Anode: a halogen if a halide ion is present, otherwise oxygen.	4	AO1 4.4.4.2
09.2	sodium is more reactive than hydrogen ; so hydrogen ions (from water) gain electrons instead of sodium ions ;		2	AO2 4.4.4.2
09.3	chlorine: bleaches damp litmus paper ; hydrogen: squeaky pop with a lit splint ; oxygen: relights a glowing splint ;		3	AO1 4.4.4.2
09.4	$4e^-$; balanced / electrons on the right-hand side (oxidation) ;		2	AO2 4.4.4.2
09.5	fill two test tubes with the solution and invert them over the electrodes ; gases collect at the top of each tube, displacing solution ; remove each tube (keeping it inverted for hydrogen) and carry out the test ;	Allow 'use a gas syringe' or 'collect over water'.	3	AO1 4.4.4.2
09.6	copper(II) ions are removed from the solution ; they gain electrons at the cathode to form copper metal ;		2	AO3 4.4.4.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.1	P: (carbon) anode ; S: (carbon) cathode / carbon lining ;		2	AO2 4.4.5
10.2	to lower the melting point (of the electrolyte) ; saving energy / reducing cost ;	Aluminium oxide melts at over 2000 °C.	2	AO2 4.4.5
10.3	cathode: aluminium ; anode: oxygen ;		2	AO1 4.4.5
10.4	oxygen is made at the anodes ; it reacts with the hot carbon to make carbon dioxide so the anodes burn away ;		2	AO2 4.4.5
10.5	$Al^{3+} + 3e^- \rightarrow Al$; $2O^{2-} \rightarrow O_2 + 4e^-$;		2	AO2 4.4.5
10.6	uses large amounts of electricity / energy ; the electrolyte has to be kept hot ;	Any one.	1	AO3 4.4.5
11.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • the ionic compound must be molten so the ions can move • positive lead ions (Pb^{2+}) move to the negative electrode (cathode) • they gain electrons: $Pb^{2+} + 2e^- \rightarrow Pb$; this is reduction; molten lead forms • negative bromide ions (Br^-) move to the positive electrode (anode) • they lose electrons: $2Br^- \rightarrow Br_2 + 2e^-$; this is oxidation; bromine gas forms (brown fumes) • electrons flow through the external circuit from the anode to the cathode		6	AO1 4.4.4.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • the reactivity series lists metals by how easily they form positive ions / react • a more reactive metal displaces a less reactive metal from its salt solution, e.g. zinc displaces copper from copper(II) sulfate • the displacement is a redox reaction: the more reactive metal loses electrons (oxidised); the less reactive metal ions gain electrons (reduced) • metals less reactive than carbon (e.g. iron, zinc, copper) are extracted by heating the ore with carbon (reduction) • metals more reactive than carbon (e.g. aluminium, sodium) are extracted by electrolysis of the molten ore • gold and other unreactive metals are found uncombined in the Earth		6	AO1 4.4.1

Where marks are lost in Topic 4.4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **OIL RIG:** Oxidation Is Loss of electrons, Reduction Is Gain. For oxygen it is the reverse idea: gaining oxygen = oxidation, losing oxygen = reduction. Say which one you mean.
- ☐ **Displacement:** name the more reactive metal and say it displaces the less reactive one. Always give a colour observation (blue solution fading, brown solid forming).
- ☐ **Salt names:** hydrochloric → chloride, sulfuric → sulfate, nitric → nitrate. The metal comes from the base, oxide or carbonate.
- ☐ **Preparing a salt:** excess insoluble solid (filter it off), then evaporate part of the water, then cool to crystallise. Use the words 'excess', 'filter', 'evaporate' and 'crystallise'.
- ☐ **Strong vs concentrated:** strong = completely ionised. Concentrated = lots of acid per dm^3. Never swap them.
- ☐ **pH change:** each pH unit is a factor of 10 in H^+ concentration. Two units = 100 times.
- ☐ **Electrolysis rules:** positive ions go to the negative cathode and are reduced (gain electrons). Negative ions go to the positive anode and are oxidised (lose electrons).
- ☐ **Aqueous electrolysis:** cathode = metal if less reactive than hydrogen, otherwise hydrogen. Anode = halogen if a halide is present, otherwise oxygen.
- ☐ **Half equations:** balance atoms and charge. Electrons go on the left for reduction and on the right for oxidation.
- ☐ **Aluminium:** cryolite lowers the melting point; the carbon anodes burn away because oxygen reacts with them. Both reasons are asked often.
- ☐ **Equations:** check that your formulae are right before you balance ($MgSO_4$, $ZnCl_2$, $Ca(OH)_2$). A wrong formula costs the balancing mark too.